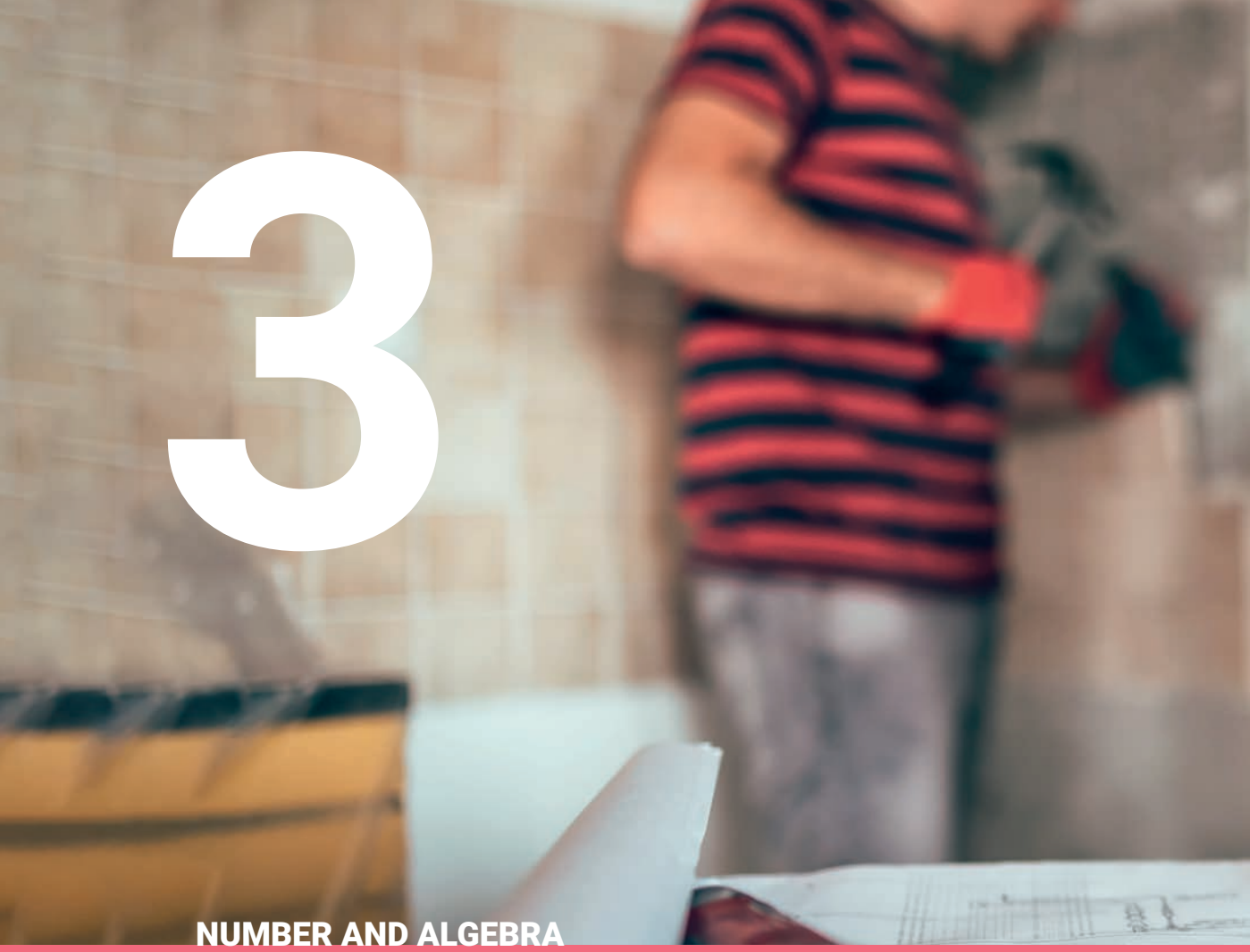


3

NUMBER AND ALGEBRA

NUMERACY AND CALCULATION

Numbers are an important part of our lives. We work with numbers when we shop for the best deal and when we borrow or invest money. Calculations are made when using cooking recipes and in the design and construction of houses. Whenever numbers are used to help us solve problems in our lives, calculations will need to be made.



Chapter outline

	Working mathematically				
3.01 Integers	U	F	PS	R	C
3.02 Decimals	U	F	PS	R	C
3.03 Fractions	U	F		R	C
3.04 Percentages	U	F			C
3.05 Applying percentages	U	F	PS	R	C
3.06 Profit, discounts and GST	U	F	PS	R	
3.07 Simple interest	U	F	PS	R	
3.08 Ratios and rates	U	F	PS	R	
3.09 Converting rates	U	F	PS	R	

Wordbank

profit The amount made when selling an item at a higher price

loss The amount lost when selling an item at a lower price

cost price The price an item costs the retailer

GST Goods and services tax, a 10% tax added to the original price of an item or service

recurring decimal A decimal with one or more digits that repeat endlessly, for example, $0.1\dot{6} = 0.1666...$

principal An amount of money invested or borrowed, on which interest is given or charged

per annum (p.a.) Per year

unitary method A method of finding a quantity by finding one part or 1% first

Quiz
Wordbank 3

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (10):

3.05 Percentage increase and decrease • The unitary method

3.06 Profit and loss • Discounts and GST • GST

3.07 Simple interest

3.08 Ratio problems • Dividing a quantity in a given ratio • Rate problems

3.09 Converting rates 2

Twig video (1):

SkillCheck Chinese development of maths

PhET interactives (3):

3.08 Ratio and proportion • Proportion playground • Unit rates

Quizzes (6):

- Wordbank 3
- SkillCheck 3
- Mental skills 3A
- Mental skills 3B
- Language of maths 3
- Test yourself 3

Skillsheets (10):

3.01 Integers • Integers using diagrams

3.02 Decimals • Multiplying by 10, 100, 1000

3.03 Improper fractions and mixed numerals • Equivalent fractions • Simplifying fractions

3.04 Fractions, decimals and percentages • Mental percentages

3.08 Ratios

Worksheets (8):

3.02 Where's the point? • Fraction families

3.05 Working with percentages

3.06 Profit and loss • Discounts and special offers

3.07 Simple interest

3.08 Rates exercise

Mind map Numeracy and calculation

Puzzles (5):

3.04 Percentages to fractions • Fractions to percentages • Percentages of an amount

3.06 Percentage profit and loss

Language of maths Numbers crossword

Technology (2):

3.04 Percentages, fractions and decimals

3.06 Profit and loss

Spreadsheet (1):

3.06 Percentages, fractions and decimals

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- add, subtract, multiply and divide integers, decimals and fractions
- convert between fractions and decimals
- solve problems involving percentages, ratios and rates
- perform calculations mentally, with pen-and-paper and using technology
- solve problems involving profit, loss, discounts, GST and simple interest
- convert between units for rates, for example, kilometres per hour to metres per second

SkillCheck ANSWERS ON P. 566

1 Evaluate each expression.

a 8×7	**b** $42 \div 6$	**c** $49 \div 7$	**d** $14 \div 2$
e 6×9	**f** $8 \div 8$	**g** 9×6	**h** 7×4
i 4×12	**j** $81 \div 9$	**k** 3×12	**l** $0 \div 8$
m $20 \div 5$	**n** $56 \div 8$	**o** 5×7	**p** $54 \div 6$

2 Simplify each fraction.

a $\frac{12}{24}$	**b** $\frac{45}{50}$	**c** $\frac{36}{80}$	**d** $\frac{21}{15}$

3 Evaluate each expression.

a $6 + 5 \times 2$	**b** $15 \div 5 + 4 \times 7$	**c** $28 \div 7 - 8 \div 2$
d $80 \div 4 \div 5$	**e** $36 \div 9 \times 5$	**f** $42 - 4 \times 8$

4 Express each number as a percentage.

a $\frac{1}{4}$	**b** 0.3	**c** 0.07	**d** $\frac{3}{5}$

Quiz
SkillCheck 3

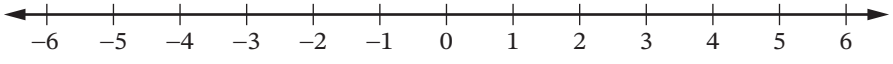
Video
Chinese development of maths

3.01

Integers 3.01

Integers are the positive and negative whole numbers and zero.

Integers can be represented on a number line as shown below.



Skillsheets
Integers

Integers using diagrams

Adding and subtracting a negative number

When **adding** a negative number, **subtract** its opposite.

When **subtracting** a negative number, **add** its opposite.

Example 1

Evaluate each expression.

a $12 + (-7)$ **b** $-3 + (-9)$ **c** $9 - (-8)$ **d** $(-12) - (-7)$

Solution

a $12 + (-7) = 12 - 7$ Adding (-7) is the same as subtracting 7
$= 5$

b $-3 + (-9) = -3 - 9$ $+ (-9) = -9$
$= -12$

c $9 - (-8) = 9 + 8$ Subtracting (-8) is the same as adding 8
$= 17$

d $-12 - (-7) = -12 + 7$ $-(-7) = +7$
$= -5$

Multiplying and dividing positive and negative numbers

- Multiplying or dividing 2 positive numbers gives a positive answer
- Multiplying or dividing a positive number by a negative number gives a negative answer
- Multiplying or dividing 2 negative numbers gives a positive answer

Example 2

Evaluate each expression.

a $-5 \times (-8)$ **b** -4×11 **c** $7 \times (-9)$

d $-72 \div (-8)$ **e** $-42 \div 6$ **f** $15 \div (-3)$

Solution

a $-5 \times (-8) = 40$ 2 negatives

b $-4 \times 11 = -44$ a negative and a positive

c $7 \times (-9) = -63$ a positive and a negative

d $-72 \div (-8) = 9$ 2 negatives

e $-42 \div 6 = -7$ a negative and a positive

f $15 \div (-3) = -5$ a positive and a negative

EXERCISE 3.01 ANSWERS ON P. 566

Integers U F PS R C

1 For each pair of integers, write > or < to make a true statement. C

a -4 ____ 7 **b** 9 ____ 12 **c** 8 ____ -3 **d** -5 ____ -8

2 Write these integers in **ascending order** (smallest to largest): 8, –13, –5, 2, –1, 0.

3 Write these integers in **descending order**: –7, –10, 2, –15, 6, –2.

EXAMPLE 1

4 Evaluate each expression.

a $8 + (-5)$ **b** $-7 + 9$ **c** $-10 + 7 + (-5)$ **d** $-5 + (-7) + 8$

e $6 - 10$ **f** $3 - (-2)$ **g** $-7 - 8$ **h** $7 - 5 - 6$

i $-9 - 11$ **j** $-8 - 5 + 4$ **k** $-10 - 3 - 5$ **l** $-12 - (-12)$

m $3 - (-2) - (-3)$ **n** $-1 - (-3) + 5$ **o** $5 - 6 - 7$ **p** $-7 + (-5) - 6$

5 The temperature at Thredbo at 6 a.m. was –12°C. By 1 p.m. the temperature had risen by 7°, but by 6 p.m., it had fallen by 5°. What is the temperature at 6:00 p.m.? PS

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6 The highest temperature recorded in Australia was 51°C at Oodnadatta, South Australia, in 1960. The highest temperature recorded in NSW was 50°C at Wilcannia in 1939. The lowest temperature recorded in Australia and NSW was –23°C at Charlotte Pass in 1994. What was the difference between the highest and lowest recorded temperatures in:



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a Australia? **b** NSW?

7 Write 2 integers that:

a have a sum of –2 **b** have a difference of –3

8 Sejuti was asked to find 2 integers less than 10 that have a difference of 15. What integers did Sejuti choose? Is there more than one answer? Give reasons for your answer. PS R

9 Evaluate each expression.

a	-3×5	**b**	$8 \times (-1)$	**c**	-4×20	**d**	$(-6) \times (-10)$
e	$(-9)^2$	**f**	$12 \times (-20)$	**g**	-8^2	**h**	-50×9
i	$45 \div (-9)$	**j**	$-80 \div (-5)$	**k**	$-24 \div 3$	**l**	$\frac{36}{-6}$
m	$\frac{-40}{-10}$	**n**	$120 \div (-6)$	**o**	$\frac{-18}{-3}$	**p**	$52 \div (-4)$

10 Evaluate each expression.

a	$7 \times (-8)$	**b**	$-120 \div (-6)$	**c**	$7 - (-15)$
d	$-10 \times 5 \times (-2)$	**e**	$\frac{27}{-3}$	**f**	$20 - 5 \times 8$
g	$(-3)^2 + (-2)^2$	**h**	$(-10 + 7) \times 4$	**i**	$-24 \div 6 \times (-5)$
j	$\frac{-100}{-5} \times 4$	**k**	$-8 - (10 - 15)$	**l**	$(-4)^3$
m	$6 - (-30) \div 6$	**n**	$40 + 8 \times (-9)$	**o**	$(-5)^2 + 4 \times (-7)$
p	$24 \div (-6) + (-5) \times 3$	**q**	$12 - (4 \times 8)$	**r**	$64 \div (-4)^2$

11 Write 2 integers that:

a have a product of –20 **b** have a quotient of –4

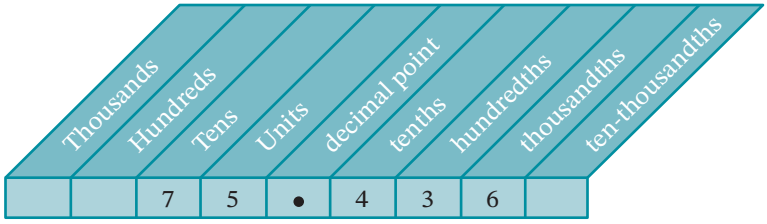
EXAMPLE 2

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3.02 Decimals

Skillsheet
Decimals

Decimals are based on powers of 10. The decimal point separates the whole number part from the fractional part.



$$75.436 = 7 \times 10 + 5 \times 1 + 4 \times \frac{1}{10} + 3 \times \frac{1}{100} + 6 \times \frac{1}{1000}$$

Example 3

Write each fraction as a decimal.

a $\frac{3}{10}$ **b** $\frac{8}{100}$ **c** $\frac{97}{1000}$

Solution

a $\frac{3}{10} = 0.3$ **b** $\frac{8}{100} = 0.08$ **c** $\frac{97}{1000} = 0.097$

Ordering decimals

To order decimals, write them so that they all have the same number of decimal places, putting in 0s where necessary, then compare the numbers.

Example 4

Arrange 35.2, 35.08 and 35.107 in ascending order.

Solution

35.107 has the most decimal places (3), so write all decimals with 3 decimal places.

35.200, 35.080, 35.107.

Then arrange the numbers in ascending order.

35.080, 35.107, 35.200.

So the correct answer is 35.08, 35.107, 35.2

Adding and subtracting decimals

When adding or subtracting decimals, keep the decimal points under one another.

Example 5

Evaluate each expression.

a $37.6 + 42.85$ **b** $327 - 156.25$

Solution

a
$$\begin{array}{r} 37.60 \\ +\ 42.85 \\ \hline 80.45 \end{array}$$

b
$$\begin{array}{r} 327.00 \\ -\ 156.25 \\ \hline 170.75 \end{array}$$

Adding 0s to give equal numbers of decimal places

Multiplying and dividing by powers of 10

- When multiplying decimals by 10, 100, 1000, ... move the decimal point 1, 2, 3, ... places to the right (the same number of places as the number of 0s in the multiplier)
- When dividing decimals by 10, 100, 1000, ... move the decimal point 1, 2, 3, ... places to the left (the same number of places as the number of 0s in the divisor)



Skillsheet
Multiplying by 10, 100, 1000

Example 6

Evaluate each product.

a 46.718×100 **b** $15.82 \div 1000$

Solution

a $46.718 \times 100 = 4671.8$ For $\times$ 100, move the point 2 places to the right

b $015.82 \div 1000 = 0.015\,82$ For $\div$ 1000, move the point 3 places to the left

Worksheet
Where's the point?

Multiplying decimals

Note: Multiplying and dividing decimals (including by whole numbers) using written methods is no longer in the syllabus. Teachers can use judgement to teach calculator methods only.

When multiplying decimals, the number of decimal places in the answer must be the same as the total number of decimal places in the question.

Example 7

Evaluate each product.

a 7.62×9 **b** 5.23×0.002

Solution

a $762 \times 9 = 6858$ Multiply without decimal points first

$7.62 \times 9 = 68.58$ The question has 2 decimal places so the answer must have 2 decimal places

b $523 \times 2 = 1046$ Multiply without decimal points first

$5.23 \times 0.002 = 0.010\,46$ The question has a total of 5 decimal places so the answer must have 5 decimal places

Dividing decimals

- When dividing by a whole number, keep the decimal points under one another.
- When dividing by another decimal, first move the decimal point in both decimals the same number of places to the right so that we are dividing by a whole number.

Example 8

Evaluate each quotient.

a $72.31 \div 4$ **b** $8.46 \div 0.005$

Solution

a
$$4\overline{)7^{3}2.31^{3}0^{2}0}\quad 18.0775$$

$72.31 \div 4 = 18.0775$

b Change the second decimal (0.005) to a whole number (5) by moving the decimal point 3 places to the right.

Then move the decimal point in the first number (8.46) 3 places to the right.

$8.460 \div 0.005 = 8460 \div 5$ Decimal point moves 3 places to the right

$$5\overline{)8460}\quad 1692$$

$8.46 \div 0.005 = 1692$

Rounding decimals

Example 9

Round 56.1835:

a to 2 decimal places

b to 3 decimal places.

Solution

a $56.1835 \approx 56.18$ — The 8 has not changed because the figure after it is 3, which is less than 5

b $56.1835 \approx 56.184$ — The 3 has been rounded up to 4 because the figure after it is 5

Recurring decimals



Worksheet
Fraction families

Recurring decimals have a pattern of numbers that repeat or recur. We use dots to indicate the digits that repeat. Here are some examples:

$0.4444... = 0.\dot{4}$

$0.588\ 88... = 0.5\dot{8}$

$0.636\ 363... = 0.\dot{6}\dot{3}$

$0.415\ 415\ 415... = 0.\dot{4}1\dot{5}$

Terminating decimals do not repeat (terminating means 'stopping'). Here are some examples:

0.6 0.875 0.057

Some decimals are neither recurring nor terminating. They have an infinite number of digits, but there is no pattern. These are **irrational numbers**. Here are some examples:

$\pi = 3.141\ 592\ 653...$ $\sqrt{3} = 1.732\ 050\ 808...$

To convert a fraction to a decimal, divide its **numerator** by its **denominator**.

Example 10

Convert each fraction to a decimal and state whether it is recurring or terminating.

a $\frac{5}{6}$

b $\frac{11}{40}$

Solution

a $\frac{5}{6} = 5 \div 6$
$= 0.8333...$
$= 0.8\dot{3}$ (a recurring decimal)

b $\frac{11}{40} = 11 \div 40$
$= 0.275$ (a terminating decimal).

EXERCISE 3.02 ANSWERS ON P. 567

Decimals U F PS R C

EXAMPLE 3

1 Write each fraction as a decimal.

a $\frac{8}{10}$ **b** $\frac{31}{100}$ **c** $\frac{456}{1000}$ **d** $\frac{57}{100}$

e $\frac{3}{100}$ **f** $\frac{92}{10000}$ **g** $\frac{6}{100}$ **h** $\frac{124}{10000}$

2 Write each decimal as a fraction.

a 0.35 **b** 0.008 **c** 0.4 **d** 0.0017

EXAMPLE 4

3 Write each set of decimals in ascending order.

a 0.7, 0.707, 7.007, 0.7007 **b** 0.202, 0.22, 0.022 02, 0.002

4 Write each set of decimals in descending order.

a 6.03, 6.1, 6.33, 0.6003, 6.3033 **b** 3.0667, 3.6, 3.66, 3.606

5 Write 3 decimals between 5.3 and 5.4 in ascending order. R C

EXAMPLE 5

6 Evaluate each expression.

a 2.34 + 5.62 **b** 8.04 + 9.761 **c** 3.458 – 2.237

d 69.24 – 41.1 **e** 34.74 – 18 **f** 24.932 + 12.05

g 15.3 – 12.582 **h** 45.804 – 27.26 **i** 231.56 + 18.407

j 6.56 + 12 **k** 32 – 16.479 **l** 45.6 – 0.007

7 Write 2 decimals that have a difference of 4.55.

8 Declan receives the following scores from seven judges for diving:

7.2 7.8 7.1 8.3 7.9 8.1 7.6

The highest and lowest scores are crossed out, and the remaining 5 scores are added to make Declan's final score. PS

a Which 2 scores are crossed out?

b What is Declan's final score?

9 **a** A piece of timber 4 metres long is cut into two. If one piece is 1.85 m, what is the length of the other piece?



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b Cate's time for swimming 100 m was 61.35 s. After training for 6 months, she reduced her time by 2.76 s. What is Cate's new time for swimming 100 m?



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c Mack ran 400 m in 59.54 s. After training for 12 months, he reduced his time to 55.6 s. By how much had Mack improved his time?

10 Evaluate each product. EXAMPLE 6

a 7.23×10	**b** 15.012×100	**c** 0.0532×1000
d 8.92×1000	**e** $35.67 \div 100$	**f** $9.7 \div 10$
g $0.987 \div 10$	**h** $234.5 \div 1000$	**i** $0.054 \div 100$

11 Evaluate each product. EXAMPLE 7

a 2.3×4	**b** 3.6×0.2	**c** 5.23×0.4
d 0.06×0.4	**e** 1.11×0.03	**f** $(0.5)^2$

12 Evaluate each quotient. EXAMPLE 8

a $34.5 \div 5$	**b** $12.08 \div 2$	**c** $1.56 \div 3$
d $12 \div 8$	**e** $4.8 \div 0.3$	**f** $8.0004 \div 0.04$

13 Evaluate each expression.

a $0.45 \times 0.2 \times 0.1$	**b** $12.6 + 0.4 \times 7$	**c** $36.9 \div 3 \times 0.02$

14 Robyn had to calculate $36.024 \div 0.08$ without using a calculator. Which of these calculations will give her the same answer? Select the correct answer **A**, **B**, **C** or **D**.

A $36\,024 \div 8$ **B** $36.024 \div 8$ **C** $3602.4 \div 8$ **D** $0.360\,24 \div 8$

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EXAMPLE 9

15 Round each decimal to the number of decimal places indicated in the brackets.

a 55.347 (2) **b** 325.75 (1) **c** 4.99 (1)

d 0.0695 (3) **e** 200.71 (0) **f** 23.8547 (3)

g 19.9 (0) **h** 46.053 (1) **i** 7.5989 (2)

j 245.67 (1) **k** 35.995 (2) **l** 301.9835 (3)

16 Which one of these decimals can be rounded to 6.75? Select **A**, **B**, **C** or **D**.

A 6.745 **B** 6.736 **C** 6.744 **D** 6.7349

17 Write each recurring decimal using dot notation. C

a 0.166 666... **b** 0.273 273 27... **c** 0.888 88... **d** 0.209 99...

18 Show the repeating pattern for each recurring decimal. C

a $0.\dot{1}09\dot{8}$ **b** $0.\dot{1}\dot{7}$ **c** $0.2\dot{5}$ **d** $8.\dot{3}1\dot{8}$

EXAMPLE 10

19 Convert each fraction to a decimal and state whether it is recurring or terminating. C

a $\frac{7}{8}$ **b** $\frac{3}{11}$ **c** $4\frac{8}{9}$ **d** $\frac{13}{5}$ **e** $\frac{19}{25}$

f $\frac{2}{3}$ **g** $\frac{13}{20}$ **h** $1\frac{7}{12}$ **i** $8\frac{13}{40}$ **j** $\frac{17}{24}$

20 **a** Convert $\frac{1}{3}$ to a recurring decimal.

b Use your answer to part **a** to determine the value of: C R

i $0.\dot{6}$ **ii** $0.\dot{9}$

21 A decimal with 3 decimal places has been rounded to 8.53. PS R

a What is the smallest decimal that can be rounded to 8.53?

b What is the largest decimal that can be rounded to 8.53?

22 Aditi and James were asked to multiply 7.28 by 5.56 and express their answers to one decimal place. Their methods are shown below.

Aditi	James
$7.28 \times 5.56 \approx 7.3 \times 5.6$ $= 40.88$ ≈ 40.9	$7.28 \times 5.56 = 40.4768$ ≈ 40.5

Discuss the methods used by both students and decide which is the more accurate answer. Give reasons. PS R C

23 Without using a calculator, evaluate:

a $(0.1)^6$ **b** $(0.2)^3$ **c** $(0.3)^4$

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Technology

Terminating and recurring decimals

Create the spreadsheet below that shows the numerators and denominators of different fractions. For example, row 2 shows $\frac{1}{2}$ and row 3 shows $\frac{3}{4}$.

J2

	A	B	C
1	**Numerator**	**Denominator**	**Decimal**
2	1	2	
3	3	4	
4	7	11	
5	3	10	
6	1	3	
7	5	6	
8	8	9	
9	12	15	
10	23	40	
11	67	99	
12	5	8	
13			

1. To calculate the equivalent decimal in column C, enter **=A2/B2** in cell C2, then **Fill Down** to cell C12.
2. In column D, classify each decimal as either T (terminating) or R (recurring).
3. Use **Format cells** for each cell in column C that has a recurring decimal and set the number of decimal places to 8.
4. Compare your answers with other students in your class.
5. Try other fractions by entering different numerators and denominators, and determine whether they convert to recurring or terminating decimals.
6. Investigate which denominators give recurring decimals by testing the unit fractions $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{20}$. Create the spreadsheet below and **Fill Down** cells A3, B3 and C2 to row 21.

	A	B	C
1	Numerator	Denominator	Decimal
2	1	2	=A2/B2
3	1	=B2+1	
4			
5			
6			

3.03 Fractions

Fractions are **rational numbers** because they can be expressed in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

Skillsheet
Improper fractions and mixed numerals

Improper fractions and mixed numerals

Example 11

a Convert $5\frac{3}{8}$ to an **improper fraction**.

b Convert $\frac{13}{4}$ to a mixed numeral.

c What whole number could be placed in the box so that $\frac{\square}{7}$ has a value between 4 and 5?

Solution

a $5\frac{3}{8} = \frac{8\times5+3}{8}$

$= \frac{43}{8}$

b $\frac{13}{4} = 13 \div 4$

$= 3\frac{1}{4}$

c $4 = \frac{4\times7}{7} = \frac{28}{7}$

$5 = \frac{5\times7}{7} = \frac{35}{7}$

$\therefore$ The number in the box must be between 28 and 35.

So the number in the box could be 29, 30, 31, 32, 33 or 34.

Ordering fractions

We can compare the size of fractions if they have a common (the same) denominator. A common denominator can be found by multiplying the denominators of the fractions together.

Skillsheet
Equivalent fractions

3.03

Example 12

a Which fraction is larger: $\frac{3}{5}$ or $\frac{2}{3}$?

b Arrange the fractions $\frac{5}{8}, \frac{2}{3}, \frac{5}{7}$ in descending order.

Solution

a A common denominator is $5 \times 3 = 15$.

Rewrite the fractions with the common denominator

$\frac{3}{5} = \frac{3 \times 3}{5 \times 3} = \frac{9}{15}$

$\frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$

$10 > 9$, so $\frac{10}{15} > \frac{9}{15}$

Compare the numerators

$\therefore \frac{2}{3}$ is the larger fraction.

b A common denominator is

$8 \times 3 \times 7 = 168$

$\frac{5}{8} = \frac{105}{168}$ ($\times 21$), $\frac{2}{3} = \frac{112}{168}$ ($\times 56$) and $\frac{5}{7} = \frac{120}{168}$ ($\times 24$)

$120 > 112 > 105$

So the fractions in descending order are $\frac{5}{7}, \frac{2}{3}, \frac{5}{8}$.

Simplifying fractions

Skillsheet
Simplifying fractions

Example 13

Simplify $\frac{27}{36}$.

Solution

$\frac{27}{36} = \frac{27 \div 9}{36 \div 9}$

Divide the numerator and denominator by a large common factor, preferably the highest common factor (HCF).

$= \frac{3}{4}$

Adding and subtracting fractions

- When **adding** or **subtracting fractions**, convert them (if needed) so that they have the same denominator.
- When **adding or subtracting mixed numerals**, add or subtract the whole numbers and fractions separately.

Example 14

Evaluate each expression.

a $\frac{3}{5}+\frac{4}{7}$ **b** $\frac{5}{6}-\frac{3}{8}$ **c** $3\frac{1}{2}+2\frac{4}{5}$ **d** $4\frac{1}{5}-2\frac{5}{8}$

Solution

a $\frac{3}{5}+\frac{4}{7}=\frac{3\times7}{35}+\frac{4\times5}{35}$ Common denominator is $5\times7=35$

$=\frac{21}{35}+\frac{20}{35}$

$=\frac{41}{35}$

$=1\frac{6}{35}$

b $\frac{5}{6}-\frac{3}{8}=\frac{5\times8}{48}-\frac{3\times6}{48}$ Common denominator is $6\times8=48$

$=\frac{40}{48}-\frac{18}{48}$

$=\frac{22\div2}{48\div2}$

$=\frac{11}{24}$

c $3\frac{1}{2}+2\frac{4}{5}=5+\frac{1}{2}+\frac{4}{5}$ Adding the whole numbers

$=5+\frac{5}{10}+\frac{8}{10}$ Common denominator is $2\times5=10$

$=5+\frac{13}{10}$

$=5+1\frac{3}{10}$ Converting the improper fraction to a mixed numeral

$=6\frac{3}{10}$

d $4\frac{1}{5}-2\frac{5}{8}=2+\frac{1}{5}-\frac{5}{8}$ Subtracting the whole numbers

$=2+\frac{8}{40}-\frac{25}{40}$ Common denominator is $5\times8=40$

$=2-\frac{17}{40}$

$=1\frac{40}{40}-\frac{17}{40}$ $2=1+1=1+\frac{40}{40}$

$=1\frac{23}{40}$

Multiplying and dividing fractions

- When **multiplying fractions**, multiply the numerators and multiply the denominators, then simplify if possible (sometimes, it is easier to simplify the fractions first).
- To **divide by a fraction** $\frac{a}{b}$, multiply by its **reciprocal** $\frac{b}{a}$.
- When **multiplying or dividing mixed numerals**, first convert them to improper fractions.

3.03

Example 15

Find:

a $\frac{2}{3}\times\frac{4}{7}$ **b** $2\frac{1}{4}\times1\frac{2}{3}$ **c** $2\times\frac{3}{5}$ **d** $\frac{3}{5}\div\frac{7}{11}$ **e** $1\frac{3}{7}\div1\frac{3}{5}$

Solution

a $\frac{2}{3}\times\frac{4}{7}=\frac{2\times4}{3\times7}$

$=\frac{8}{21}$

Multiplying numerators and multiplying denominators

b $2\frac{1}{4}\times1\frac{2}{3}=\frac{9}{4}\times\frac{5}{3}$

$=\frac{45}{12}$

$=3\frac{9}{12}$

$=3\frac{3}{4}$

Convert to improper fractions

c $2\times\frac{3}{5}=\frac{2}{1}\times\frac{3}{5}$

$=\frac{2\times3}{1\times5}$

$=\frac{6}{5}$

$=1\frac{1}{5}$

Write 2 as a fraction $\frac{2}{1}$ first

d $\frac{3}{5}\div\frac{7}{11}=\frac{3}{5}\times\frac{11}{7}$

$=\frac{33}{35}$

Multiply by the reciprocal of $\frac{7}{11}$

e $1\frac{3}{7}\div1\frac{3}{5}=\frac{10}{7}\div\frac{8}{5}$

$=\frac{10}{7}\times\frac{5}{8}$

$=\frac{50}{56}$

$=\frac{25}{28}$

Convert to improper fractions

EXERCISE 3.03 ANSWERS ON P. 567

Fractions U F R C

EXAMPLE 11

1 Convert each mixed numeral to an improper fraction.

a $1\frac{2}{3}$ **b** $4\frac{1}{2}$ **c** $2\frac{3}{5}$ **d** $5\frac{11}{20}$

2 Convert each improper fraction to a mixed numeral.

a $\frac{7}{3}$ **b** $\frac{11}{7}$ **c** $\frac{16}{5}$ **d** $\frac{37}{10}$

EXAMPLE 12

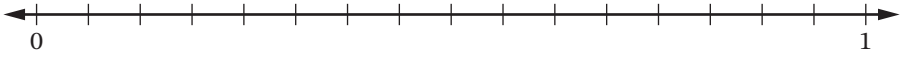
3 For each pair of fractions, determine which is larger.

a $\frac{5}{6}, \frac{7}{8}$ **b** $\frac{7}{10}, \frac{3}{4}$ **c** $\frac{7}{8}, \frac{9}{11}$

4 Write each set of fractions in descending order.

a $\frac{3}{5}, \frac{3}{4}, \frac{6}{7}$ **b** $\frac{5}{11}, \frac{1}{2}, \frac{2}{3}$ **c** $\frac{4}{5}, \frac{7}{8}, \frac{17}{20}$

5 Copy the number line below and clearly mark the positions of the fractions $\frac{1}{3}, \frac{4}{5}, \frac{3}{4}, \frac{1}{2}$.



EXAMPLE 13

6 Simplify each fraction.

a $\frac{18}{27}$ **b** $\frac{48}{64}$ **c** $\frac{15}{33}$ **d** $\frac{28}{35}$

EXAMPLE 14

7 Evaluate each expression. Answer in simplest form.

a $\frac{2}{3}+\frac{4}{7}$ **b** $\frac{3}{5}+\frac{1}{2}$ **c** $\frac{5}{9}-\frac{1}{4}$ **d** $\frac{9}{10}-\frac{2}{3}$

e $\frac{3}{8}+\frac{7}{10}$ **f** $\frac{5}{6}-\frac{3}{4}$ **g** $\frac{5}{11}-\frac{1}{3}$ **h** $\frac{17}{25}-\frac{1}{4}$

i $2\frac{3}{4}+1\frac{2}{5}$ **j** $3\frac{7}{10}+2\frac{1}{3}$ **k** $2\frac{4}{5}-1\frac{2}{3}$ **l** $4\frac{3}{7}-1\frac{4}{5}$

EXAMPLE 15

8 Evaluate each product.

a $\frac{5}{7}\times\frac{3}{4}$ **b** $\frac{1}{2}\times\frac{3}{5}$ **c** $6\times\frac{3}{4}$ **d** $\frac{2}{3}\times 7$

e $\frac{7}{8}\times\frac{2}{3}$ **f** $\frac{7}{8}\times 24$ **g** $\frac{1}{4}\times\frac{7}{9}$ **h** $\frac{4}{5}\times 8$

i $\frac{4}{5}$ of 10 m **j** $\frac{3}{8}$ of 40 L **k** $\frac{7}{10}$ of \$500 **l** $\frac{3}{4}$ of 76 kg

m $1\frac{1}{2}\times 3\frac{5}{6}$ **n** $5\frac{2}{3}\times 2\frac{3}{5}$ **o** $\frac{5}{6}\times 1\frac{1}{5}$ **p** $\left(2\frac{1}{2}\right)^2$

9 Evaluate each quotient.

a $\frac{3}{4}\div\frac{1}{2}$ **b** $\frac{2}{3}\div\frac{4}{5}$ **c** $\frac{3}{7}\div 5$ **d** $\frac{6}{7}\div\frac{2}{3}$

e $2\frac{1}{2}\div 4$ **f** $1\frac{7}{8}\div 2\frac{2}{3}$ **g** $3\frac{3}{4}\div 1\frac{1}{5}$ **h** $5\frac{1}{2}\div 2\frac{1}{4}$

10 Write a fraction between $\frac{3}{5}$ and $\frac{2}{3}$.

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EXAMPLE 11

3.03

11 For each improper fraction below, select the correct number **A**, **B**, **C** or **D** that should go in the box. R

a $\frac{\square}{5}$ has a value between 7 and 8.

A 40 **B** 37 **C** 35 **D** 33

b $\frac{\square}{9}$ has a value between 4 and 5.

A 30 **B** 35 **C** 40 **D** 45

c $\frac{35}{\square}$ has a value between 3 and 4.

A 6 **B** 8 **C** 10 **D** 12

12 The sum of 2 numbers is $5\frac{1}{2}$. If one of the numbers is $2\frac{3}{5}$, what is the other number?

13 At a football match, 14 players are given three-quarters of an orange to eat. How many oranges are eaten? PS

14 Taylor worked $5\frac{1}{2}$ hours yesterday and $\frac{1}{2}$ of that today. How long did Taylor work today?

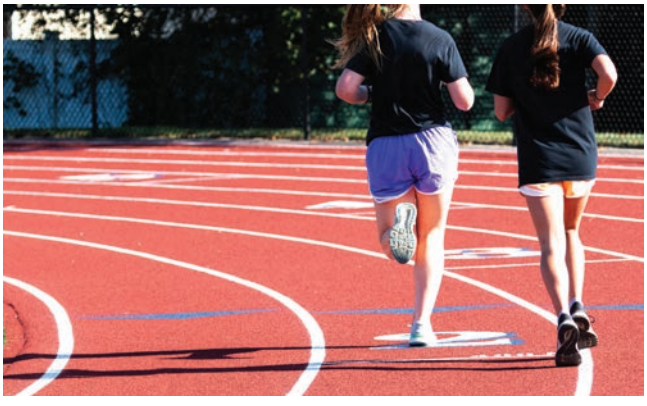
15 A driver education course requires 20 hours of classroom instruction. How many lessons are required if each lesson lasts $1\frac{1}{2}$ hours? PS

16 Cassie spent $\frac{1}{3}$ of the day working, $\frac{5}{12}$ of the day sleeping, $\frac{1}{8}$ of the day travelling and the rest of the day relaxing. What **fraction** of the day did she spend relaxing?

17 Find 2 mixed numerals that have: R

a a sum of $10\frac{1}{2}$ **b** a product of $10\frac{1}{2}$

18 At Nelson College's athletic carnival, $\frac{7}{9}$ of the students attended. There are 1188 students enrolled at Nelson College. How many students attended the carnival?



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19 Chloe ran $4\frac{1}{2}$ laps of an oval in 20 minutes. How long did it take her to run one lap?

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Did you know?

Diophantus of Alexandria

Diophantus of Alexandria is known as the 'Father of Higher Arithmetic'. He was an outstanding Greek mathematician who, along with Pappus, dominated Greek mathematics from 250 CE to 350 CE. Diophantus introduced some algebraic symbolism into mathematics, and collected and catalogued what the Greeks had achieved in algebra.

Not much is known about the life of Diophantus. However, part of what is known comes from a problem that was inscribed on his tombstone, as shown.

What was Diophantus' age when he died?



DIOPHANTUS PASSED $\frac{1}{6}$ OF HIS LIFE IN CHILDHOOD, $\frac{1}{12}$ IN YOUTH, AND $\frac{1}{7}$ MORE AS A BACHELOR. FIVE YEARS AFTER HIS MARRIAGE WAS BORN A SON WHO DIED FOUR YEARS BEFORE HIS FATHER AT HALF THE AGE AT WHICH HIS FATHER DIED.



Quiz
Mental skills 3A

Mental skills 3A: Maths without calculators ANSWERS ON P. 567

Squaring a number ending in 5, 1 or 9

The square of a number ending in 5 always ends in 25.

For example, $35^2 = 1225$ and $105^2 = 11\,025$.

A mental calculation trick requires three easy steps:

- delete the 5 from the number
- multiply the remaining number by the next consecutive number
- write '25' at the end of the product.

1 Study each example.

a To calculate 35^2:

- deleting the 5 from 35 leaves 3
- multiply 3 by the next consecutive number: $3 \times 4 = 12$
- write '25' at the end: 1225

$35^2 = 1225$

b To calculate 105^2:

- deleting the 5 from 105 leaves 10
- multiply 10 by the next consecutive number: $10 \times 11 = 110$
- write 25 at the end: 11 025

$105^2 = 11\,025$

2 Now calculate each square number.

a	25^2	**b**	55^2	**c**	45^2	**d**	65^2
e	75^2	**f**	2.5^2	**g**	4.5^2	**h**	205^2
i	6.5^2	**j**	7.5^2	**k**	275^2	**l**	345^2

The square of a number ending in 1 always ends in 1.

For example, $41^2 = 1681$ and $71^2 = 5041$.

A mental calculation trick requires three steps.

- Round the number to the nearest 10 (by subtracting 1) to make a new number
- Square the new number
- Add to your answer the new number and its next consecutive number

3 Study each example.

a To calculate 41^2:

- Round 41 down to 40
- Square 40: $40^2 = 1600$
- Add 40 and 41: $1600 + 40 + 41 = 1681$

$41^2 = 1681$

b To calculate 71^2:

- $70^2 = 4900$ Rounding down and squaring
- $4900 + 70 + 71 = 5041$

$71^2 = 5041$

4 Now calculate each square number.

a	21^2	**b**	101^2	**c**	31^2	**d**	91^2	**e**	5.1^2
f	81^2	**g**	61^2	**h**	201^2	**i**	2.1^2	**j**	9.1^2

The square of a number ending in 9 also ends in 1.

For example, $29^2 = 841$ and $99^2 = 9801$.

A mental calculation trick requires three steps:

- round the number to the nearest 10 (by adding 1) to make a new number
- square the new number
- subtract from your answer the new number and its *previous* consecutive number

5 Study each example.

a To calculate 29^2:

- round up to 30
- square 30: $30^2 = 900$
- subtract 30 and 29: $900 - 30 - 29 = 841$

$29^2 = 841$

b To calculate 99^2:

- round up to 100
- square 100: $100^2 = 10\,000$
- $10\,000 - 100 - 99 = 9801$

$99^2 = 9801$

6 Now calculate each square number.

a	59^2	**b**	69^2	**c**	89^2	**d**	19^2	**e**	109^2
f	5.9^2	**g**	6.9^2	**h**	119^2	**i**	39^2	**j**	8.9^2

3.04 Percentages

A **percentage** is a fraction 'out of 100' (denominator of 100). For example, 15%, means '15 out of 100' or $\frac{15}{100}$ (which can be simplified to $\frac{3}{20}$).

Skillsheets
Fractions, decimals and percentages

Mental percentages

Puzzles
Percentages to fractions

Fractions to percentages

Technology
Percentages, fractions and decimals

Spreadsheet
Percentages, fractions and decimals

Converting from percentages

- To **convert a percentage to a fraction**, write it as a fraction with denominator 100 and simplify
- To **convert a percentage to a decimal**, divide by 100

Example 16

Convert each percentage to a simplified fraction.

a 24% **b** $5\frac{1}{2}\%$ **c** 8.2%

Solution

a $24\% = \frac{24 \div 4}{100 \div 4}$
$= \frac{6}{25}$

b $5\frac{1}{2}\% = \frac{5\frac{1}{2}}{100}$
$= \frac{5\frac{1}{2} \times 2}{100 \times 2}$
$= \frac{11}{200}$

c $8.2\% = \frac{8.2}{100}$
$= \frac{8.2 \times 10}{100 \times 10}$
$= \frac{82}{1000}$
$= \frac{41}{500}$

Example 17

Convert each percentage to a decimal.

a 18% **b** 3.5% **c** 215%

Solution

a $18\% = 18 \div 100$
$= 0.18$

b $3.5\% = 3.5 \div 100$
$= 0.035$

c $215\% = 215 \div 100$
$= 2.15$

Converting to percentages

To **convert a fraction or a decimal to a percentage**, multiply by 100%.

Example 18

Convert each fraction or decimal to a percentage.

a $\frac{5}{8}$ **b** 0.085 **c** 0.3 **d** $\frac{5}{6}$

Solution

a $\frac{5}{8} = \frac{5}{8} \times 100\%$

$= 62\frac{1}{2}\%$ (or 62.5%)

b $0.085 = 0.085 \times 100\%$

$= 8.5\%$

c $0.3 = 0.30 \times 100\%$

$= 30\%$

d $\frac{5}{6} = \frac{5}{6} \times 100\%$

$= 83\frac{1}{3}\%$ (or $83.\dot{3}\%$)

Ordering fractions, decimals and percentages

Example 19

Arrange $\frac{3}{5}$, 5%, 0.5 and $\frac{5}{9}$ in ascending order.

Solution

Convert each number to a percentage or decimal.

Method 1: as percentages

$\frac{3}{5} = 60\%$, $5\% = 5\%$, $0.5 = 50\%$, $\frac{5}{9} = 55.\dot{5}\%$

In ascending order:

$5\%, 50\%, 55.\dot{5}\%, 60\%$

$\therefore$ The ascending order is: $5\%, 0.5, \frac{5}{9}, \frac{3}{5}$.

Method 2: as decimals

$\frac{3}{5} = 0.6$, $5\% = 0.05$, $0.5 = 0.5$, $\frac{5}{9} = 0.\dot{5}$

In ascending order:

$0.05, 0.50, 0.\dot{5}, 0.60$

Percentage of a quantity

Example 20

Find:

a 12% of \$90 **b** $5\frac{1}{2}\%$ of \$540

Solution

a $12\% \times \$90 = \frac{12}{100} \times \$90$ or $0.12 \times \$90$

$= \$10.80$

Entering the decimal 0.12 for 12% on a calculator is faster

b $5\frac{1}{2}\%$ of $\$540 = 5\frac{1}{2} \div 100 \times \$540$ or $0.055 \times \$540 = \$29.70$

3.04

Puzzle
Percentages of an amount

EXERCISE 3.04 ANSWERS ON P. 568

Percentages U F C

EXAMPLE 16

1 Convert each percentage to a simplified fraction. C

a 6%	**b** 45%	**c** 84%	**d** 2%
e 0.5%	**f** 150%	**g** 51%	**h** $4\frac{1}{2}\%$
i $8\frac{3}{4}\%$	**j** $33\frac{1}{3}\%$	**k** $12\frac{1}{2}\%$	**l** 4.75%
m 6.25%	**n** 5.5%	**o** 30%	**p** $16\frac{2}{3}\%$

EXAMPLE 17

2 Convert each percentage to a decimal. C

a 85%	**b** 5.1%	**c** 10%	**d** 8.45%
e $6\frac{1}{2}\%$	**f** 12.25%	**g** 0.4%	**h** 450%

EXAMPLE 18

3 Convert each fraction or decimal to a percentage. C

a $\frac{7}{100}$	**b** $2\frac{3}{8}$	**c** 0.48	**d** 0.055
e $\frac{36}{50}$	**f** 0.016	**g** $\frac{2}{3}$	**h** $\frac{3}{4}$
i $\frac{11}{25}$	**j** 0.73	**k** $\frac{3}{15}$	**l** $\frac{7}{11}$
m 1.015	**n** $\frac{1}{6}$	**o** $\frac{18}{20}$	**p** 0.9

EXAMPLE 19

4 State whether each statement is true (T) or false (F).

a $55\% > 0.58$	**b** $\frac{6}{11} < 55\%$	**c** $24.5\% > \frac{2}{9}$	**d** $0.87 < \frac{7}{8}$

5 Write each set of numbers in ascending order:

a $46\%, \frac{3}{7}, 0.427, \frac{2}{5}$ **b** $\frac{29}{40}, 70\%, 0.73, \frac{8}{11}$

6 Write each set of numbers in descending order:

a $0.333, \frac{4}{11}, 33.33\%, \frac{1}{3}$ **b** $\frac{4}{7}, 55\%, 0.58, \frac{1}{2}$

EXAMPLE 20

7 Find:

a 15% of \$85	**b** 120% of 60 kg	**c** 45% of 16 m
d 2.4% of \$350	**e** $5\frac{1}{4}\%$ of \$80	**f** 12% of 95 kg
g $2\frac{1}{2}\%$ of \$870	**h** 150% of 16 L	**i** 3.75% of \$2850
j $33\frac{1}{3}\%$ of \$6540	**k** 6.25% of 25 t	**l** 0.4% of \$3720

8 A total of 84 600 people attended the State of Origin match between NSW and Queensland in Sydney. If 71% of the crowd supported NSW:

a what percentage of the crowd supported Queensland?

b how many people in the crowd supported Queensland?



Newspix/Brett Costello

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9 980 students attend St Klaas' College. If 19% of them are in Year 9, calculate the number of Year 9 students.

10 Buzz Coffee advertises an extra 20% of coffee for the same price. If the normal amount is 700 g, how much extra coffee will you get for the same price?

11 Jakub's weight is now 125% of what it was a year ago. If his weight then was 85 kg, what is his weight now?

Did you know?

Blood: it takes all types

The blood groups in Australia are as follows.

O		A		B		AB	
+	−	+	−	+	−	+	−
40%	9%	31%	7%	8%	2%	2%	1%

1 If Australia's population is 26 million, how many people would be in each blood group?

2 Do you know what your blood group is? How would you find out?



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Technology

Converting fractions to percentages

In this activity, we will use a spreadsheet to convert 12 fractions to percentages and sort them in descending order.

1 Create a spreadsheet like the one at right that shows the numerators and denominators of different fractions. For example, row 2 shows $\frac{1}{4}$ and row 3 shows $\frac{3}{5}$.

2 Enter **=A2/B2*100** in cell C2 to convert $\frac{1}{4}$ to a **decimal** first.

3 **Fill Down** from cell C2 to C13 to convert each fraction to a decimal.

4 Use **Format cells** to set column C in **percentage** format to 5 decimal places: this converts all the decimals to percentages.

	A	B	C
1	Numerator	Denominator	Percentage
2	1	4	
3	3	5	
4	9	10	
5	7	8	
6	2	3	
7	28	50	
8	45	75	
9	5	6	
10	27	60	
11	76	80	
12	7	9	
13	375	1000	
14			

5 Highlight rows 2 to 13 and click **Sort** column C by **descending (largest to smallest)**.

6 Try converting different fractions to percentages and sorting them.

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3.05 Applying percentages

Video
Percentage increase and decrease

Worksheet
Working with percentages

Example 21

a Increase $200 by 7%.

b Decrease $150 by 12%.

Solution

a Increase $= 7\% \times \$200$
$= \$14$
$\therefore$ New amount $= \$200 + \$14$
$= \$214$ (100% + 7% = 107%)
or New amount $= 107\%$ of $200
$= 1.07 \times \$200$
$= \$214$

b Decrease $= 12\% \times \$150$
$= \$18$
$\therefore$ New amount $= \$150 - \$18$
$= \$132$
or New amount $= 88\%$ of $150
(100% − 12% = 88%) $= 0.88 \times \$150$
$= \$132$

Expressing quantities as percentages

To **express one quantity as a percentage of another**, write the appropriate fraction and then change it to a percentage:

$$\text{Percentage} = \frac{\text{amount}}{\text{whole amount}} \times 100\%$$

or Percentage = amount ÷ whole amount × 100%

Example 22

a Jamila scores 43 out of 60 in a History test. What is her percentage mark to the nearest whole number?

b What percentage (correct to one decimal place) is 35 s of 5 min?

Solution

a Percentage mark $= \frac{43}{60} \times 100\%$
$= 71.666...\%$
$\approx 72\%$

Use $\text{Percentage} = \frac{\text{amount}}{\text{whole amount}} \times 100\%$

b Percentage $= \frac{35\text{ s}}{5\text{ min}} \times 100\%$
$= \frac{35\text{ s}}{300\text{ s}} \times 100\%$
$= 11.666...\%$
$\approx 11.7\%$

Change minutes to seconds so that the same units are being used

5 min = 5 × 60 s = 300 s

The **unitary method** can be used to find the whole amount (100%) when part of the amount is known. 'Unit' means 'one' and the unitary method involves finding 1% first.

The unitary method

Given a percentage of an amount, to find the whole amount:

1 find 1% of the amount by dividing by the known percentage

2 multiply by 100 to find the whole amount (100%).

Video
The unitary method

3.05

Example 23

This week, Julio received a bonus of $86, which was 9% of his weekly wage. Calculate Julio's weekly wage, correct to the nearest dollar.

Solution

9% of the wage = $86

$\therefore$ 1% of the wage = $86 ÷ 9

= $9.555...

$\therefore$ Julio's wage = $9.55... × 100 100% of the wage

= $955.555...

≈ $956

EXERCISE 3.05 ANSWERS ON P. 568

Applying percentages U F PS R C

EXAMPLE 21

1 Increase:

a $455 by 10% **b** $80 by 15.5% **c** $680 by 200%

d 80 kg by 2.5% **e** 35 m by 30% **f** 120 L by 140%

2 Decrease:

a $830 by 35% **b** 120 min by 5% **c** 90 L by 12%

d 6.8 m by 25% **e** 130 kg by 18% **f** $96.50 by 10%

3 Matt buys a bike for $860 and sells it, making a profit of 5%.

a How much profit did Matt make?

b For what price did Matt sell the bike?

4 T-shirts priced at $38 are reduced by 45% at an end-of-summer sale. What is the sale price of the T-shirts?

5 Due to severe storms in summer, the price of fresh fruit and vegetables increased by 34%. How much will broccoli cost if its original price was $4.99/kg?

6 Lucy's weekly wage of $895 increased by 5.5%. Find her new weekly wage.

7 A car dealer is offering new year sales. What will you pay for a used car marked at $16 500 if a **discount** of 12% is given?

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EXAMPLE 22

8 Express each proportion as a percentage, correct to one decimal place if necessary. C

a	55 out of 80	**b**	80 out of 120	**c**	45 out of 70
d	15 out of 18	**e**	$8 out of $12	**f**	$9 out of $60
g	35 s out of 4 min	**h**	15 min out of 2 h	**i**	340 mm out of 2 m
j	3 hours out of 1 day	**k**	85 cm out of 5 m	**l**	140c out of $10

9 Selina achieved the following marks in 2 tests: PS R C

English: 32 out of 40

Mathematics: 47 out of 60

Convert each mark to a percentage to decide which is the better mark.

10 Travis earns $840 per week, of which $280 goes to rent. Find what percentage of Travis' wage goes to rent, correct to one decimal place.

11 Bozena earns $480 per week while she is completing Year 12 at school. If she spends $95 of it on petrol, what percentage of her wage is this, correct to one decimal place?

12 The nutritional information on a food label is shown. C

a What percentage of a 100 g serve is:

i protein?

ii carbohydrate?

b For a 100 g serve, what percentage of the total fat is saturated fat?

	Per 100 g
Energy	2105 kJ
Total Fat (of which is Saturated fat)	25 g 16 g
Carbohydrate (of which is sugars)	62 g 38 g
Protein	5 g
Salt	1 g

EXAMPLE 23

13 In the Netball World Cup final, Australia scored 15 goals in the second quarter. If this represented 29.4% of the goals Australia scored in the match, how many goals did Australia score in total? PS R

14 A jeweller charges $105 for valuing a diamond ring. If this charge represents 1.25% of the value of the ring, find this value correct to the nearest dollar.

15 A can of energy food drink contains 20% more for the same price as the standard can. If the extra amounts to 150 g, how much does the standard can contain?

16 Last year, Ben paid 28% of his salary in tax. If he paid $23 423 in tax, what was Ben's salary? Answer to the nearest dollar.

17 An amount is increased by 10% and then decreased by 10%. What is the overall percentage change? PS R

18 An amount is decreased by 10% and then increased by 10%. What is the overall percentage change? PS R

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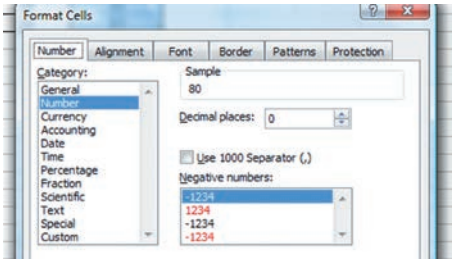
Technology

Converting marks to percentages

In this activity, you will convert students' test marks from a score out of 60 to a percentage.

1 Create the spreadsheet shown below and enter the formula shown in C3. Use it to convert each mark out of 60 to a percentage.

	A	B	C
1	**Student**	**Mark**	**Percentage**
2		60	%
3	Sam	48	=B3/B2*100
4	Tranh	37	
5	Mithra	59	
6	Anka	47	
7	Serg	51	
8	Rebecca	39	
9	Suriya	50	
10			



2 Highlight column C and choose **Number** and set the decimal places to 0.

3 Use your spreadsheet to answer these questions. Write your answers in the cells indicated in brackets [ ].

- **a** Who scored the highest mark? [D1]
- **b** What mark out of 60 is equivalent to 80%? [D2]
- **c** How many students scored between 70% and 90%? [D3]
- **d** How many students scored less than 70%? [D4]
- **e** What was the difference between the highest and lowest percentage marks? [D5]

4 Select columns 3 to 9 and sort the marks in descending order.

Profit, discounts and GST 3.06

Profit and loss

- When an item is sold for **more** than it costs, a **profit** is made.
- When an item is sold for **less** than it costs, a **loss** is made.
- The percentage profit and loss are usually calculated as a percentage of the **cost price**.

Worksheet
Profit and loss

Puzzle
Percentage profit or loss

Example 24

The online store Jackson's buys notebook computers for $550 each and sells them for $680. Find:

a the profit on each computer

b the percentage profit on the cost price, correct to one decimal place.

Solution

a Profit = $680 − $550
= $130

b Percentage profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$
$= \frac{130}{550} \times 100\%$
$= 23.6363...$
$\approx 23.6\%$

Discounts



Video
Discounts and GST



Worksheet
Discounts and special offers

Example 25

At *Jeans Minus*' end-of-season sale, jeans are discounted by 25%. What is the discount price of a pair of jeans marked at $148?

Solution

Method 1

Discount = 25% of $148
= $37
Discount price = $148 − $37
= $111

Method 2

Discount price = 75% of $148
= $111



Video
Discounts and GST

Example 26

A home cinema costs $865 after a 24% discount. What was its original price, rounded to the nearest dollar?

Solution

Sale price = $865, discount = 24%

∴ 76% of the original price = $865 — 100% − 24% = 76%

$1\% = \frac{\$865}{76}$ — Using the unitary method
$= \$11.381\,578...$

∴ Original price = $11.381 578... × 100 — To find 100%
= $1138.157...
≈ $1138

GST

Goods and Services Tax (GST) is a tax paid to the government on most goods (items) and services that we purchase. In Australia, GST is charged at 10% of the original price and is generally included in the marked price of the good or service.

Example 27

The **selling price** of a lounge suite is $2695 with GST included. How much of this price is the GST?

Solution

Selling price = 100% + 10% Adding 10% GST

= 110%

110% of the original price = $2695

10% GST = $2695 ÷ 11

= $245

Videos
Discounts and GST
GST

3.06

EXERCISE 3.06 ANSWERS ON P. 568

Profits, discounts and GST U F PS R

In this exercise, express all percentages correct to one decimal place and all prices to the nearest cent.

EXAMPLE 24

1 A home theatre system with a cost price of $650 is sold for $1250. Find:

a the profit **b** the percentage profit on the cost price.

2 A motorbike originally priced at $16 000 is sold for $14 500. Find:

a the loss **b** the percentage loss

3 Alla bought an old car for $3000, spent another $1500 on repairs and then sold it for $6200. Find:

a the total amount that Alla spent on the car

b the profit made when she sold the car

c her percentage profit.

4 A house was bought for $850 000. A year later, it was sold for $720 000. Calculate:

a the loss **b** the percentage loss

5 Towels with a cost price of $25 each are sold for $35. Calculate the percentage profit.

6 A car costing $12 000 is sold for $8000. Calculate the percentage loss.

7 *Home Necessities* store buys cutlery sets for $120 that are then sold for $230. Calculate the percentage profit.

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EXAMPLE 25

8 GIA supermarket reduces the price of some items by 35%. Find the discount price of the following items given their original prices.

a 250 g butter, $2.40 **b** 2 L milk, $3

c 1 kg cheese, $11 **d** 24 cans of soft drink, $15.99

9 Calculate the selling price of a computer marked at $760, now discounted by 15%. Select the correct answer **A**, **B**, **C** or **D**.

A $114 **B** $660 **C** $646 **D** $636

10 The price of a punnet of strawberries marked at $4.95 was reduced by 25%. Calculate its discounted price. Select **A**, **B**, **C** or **D**.

A $4.74 **B** $3.74 **C** $3.71 **D** $1.25

11 At a book sale, all books are discounted by 35%. What is the discount price of a book marked at $25.50?

12 A factory-seconds washing machine has its price reduced from $980 to $750. What is the percentage discount?

13 A TV priced at $1599 is sold for $1250. Find the discount as a percentage of the marked price.

14 Calculate the percentage discount for each item.

a



b



EXAMPLE 26

15 A jacket discounted by 40% after Christmas sells for $84. What was the original price of the jacket?

16 A bedroom suite reduced by 15% is sold for $2801. What was its original price? Answer to the nearest dollar.

17 Running shoes are sold to make a profit of 85%. If they sold for $164.65, what was their cost price? Select **A**, **B**, **C** or **D**.

A $140 **B** $89 **C** $76 **D** $65

18 A plane fare from Brisbane to Sydney has been discounted by 45%. If Amy paid $159 for a discounted ticket, what was the full price to the nearest dollar?

19 For each item, calculate the GST payable and the final price.

a a car priced at $20 900 **b** a home cinema priced at $1810

c a theatre ticket priced at $145 **d** plumber's fees of $180

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EXAMPLE 27

3.06

20 Given the selling price of each item with GST included, calculate the GST and the original price, correct to the nearest cent. PS R

a haircut: $46.20
b car repairs: $739.20
c photo frame: $31.35
d pair of jeans: $65.89
e refrigerator: $924
f piano lessons: $198 per term



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21 A discount warehouse has a 30% discount on all computers with a further 5% off the discount price if you pay in cash. Carlos wants to buy a computer priced at $1899. PS R

a What is the discount price of the computer?
b Carlos pays by cash. How much will he need to pay now, to the nearest 5 cents?
c What is the total discount Carlos received?
d Calculate his total percentage discount (on the original cost).

22 Justine buys 20 boxes of paper at $29.95 a box. PS R

a What is the cost of the 20 boxes of paper?
b Justine receives a business discount of 15%. How much will she need to pay?
c Justine pays by cash and receives a further discount of 5%. How much will she now pay, to the nearest 5 cents?
d Find the amount Justine would pay if she received a single discount of 20% on the cost of the paper.
e Is the single discount of 20% the same as the 2 successive discounts of 15% and 5%? Give reasons for your answer.

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Technology

Profit or loss

The spreadsheet below shows items for sale in *Nina's Furniture Warehouse*. Each item is shown with its cost price and selling price. Devesh and Sarita want to buy these items for their new family home.

1 Create a new spreadsheet as shown.

	A	B	C	D	E
1	**Item**	**Cost price**	**Selling price**	**Profit or Loss**	**% Profit or Loss**
2	lounge	599	799		
3	bookshelf	189	169		
4	childrens table	110	95		
5	child's cot	350	475		
6	armchair	239	310		
7	rug	695	490		
8	mirror	45	79		
9	dining table	865	1060		
10					

2 Highlight columns B, C and D and choose **Format cells** and **Currency**.

3 In cell D2, enter the formula **=C2 – B2** to calculate the profit or loss for the lounge (A loss will be shown as a negative value).

4 Use **Fill down** in column D to calculate the profit or loss for each item.

5 In cell E2, enter **=D2/B2*100** to calculate the percentage profit or loss for the lounge.

6 Use **Fill down** in column E.

7 Highlight column E and choose **Format cells** and **Number** and set the number of decimal places to 0.

8 How much money did Devesh and Sarita spend altogether on furniture? In cell C10, enter **=sum(C2:C9)**.

9 Calculate the total profit or loss made by the store on these items. In cell D10, enter **=sum(D2:D9)**.



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Simple interest 3.07

Banks, credit unions and other financial institutions reward investors by paying them interest on their savings or investments. Conversely, they charge borrowers by making them pay interest on their loans.

- The original amount of money invested or borrowed is called the **principal**.
- **Interest** is calculated as a percentage of the principal.
- This percentage is called the **interest rate**, usually written as a rate **per annum** ('per year'), abbreviated 'p.a.'.
- **Simple interest** (or **flat rate interest**) is interest calculated simply on the original principal.

Worksheet Simple interest 3.07

Example 28

Find the simple interest earned on an investment of $18 000 at 3.5% p.a. for 5 years.

Solution

Principal = $18 000, interest rate = 3.5% p.a., term = 5 years.

Term means the amount of time the investment or loan has occurred

Interest = 3.5% × $18 000 × 5 Over 5 years

= $3150

The simple interest formula

$I = Prn$, where:

I is the simple interest

P is the principal

r is the interest rate per time period, expressed as a decimal

n is the number of time periods

Applying this to the above example,

P = $18 000, r = 3.5% = 0.035, n = 5 years

$I = Prn$

= $18 000 × 0.035 × 5

= $3150



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Video
Simple interest

Example 29

Find the simple interest on:

a \$8620 at 2.4% p.a. for 7 months

b \$5600 at 6.25% p.a. for 220 days

Solution

a $P = \$8620$, $r = 2.4\% = 0.024$,

$n = 7 \text{ months} = \frac{7}{12} \text{ years}$ — n must be in years as the interest rate is per year

$I = Prn$

$= \$8620 \times 0.024 \times \frac{7}{12}$

$= \$120.68$

b $P = \$5600$, $r = 6.25\% = 0.0625$,

$n = 220 \text{ days} = \frac{220}{365} \text{ years}$

$I = Prn$

$= \$5600 \times 0.0625 \times \frac{220}{365}$

$= \$210.9589...$

$\approx \$210.96$ — rounded to the nearest cent

Video
Simple interest

Example 30

After 2 years, an investment of \$1560 has earned \$87.36 in simple interest.

What is the annual interest rate?

Solution

$I = \$87.36$, $P = \$1560$, $n = 2$ years

$I = Prn$

$\$87.36 = \$1560 \times r \times 2$

$\$87.36 = \$3120r$

$r = \frac{\$87.36}{\$3120}$

$= 0.028$

$= 2.8\%$

$\therefore$ Annual interest rate $= 2.8\%$

Example 31

For how long will \$12 000 need to be invested to earn \$1350 if the flat rate of interest is 4.5% p.a.?

Solution

$I = \$1350$, $P = \$12\,000$, $n = 4.5\% = 0.045$

$I = Prn$

$1350 = 12\,000 \times 0.045 \times \text{n}$

$1350 = 540n$

$n = \frac{1350}{540}$

$= 2.5$ years

EXERCISE 3.07 ANSWERS ON P. 568

Simple interest U F PS R

In this exercise, give all money answers correct to the nearest cent.

EXAMPLE 28

1 Calculate the simple interest earned on each investment.

a $18 000 at 7% p.a. for 5 years
b $5280 at 2.85% p.a. for 2 years
c $3000 at 5.5% p.a. for 4 years
d $9450 at $2\frac{1}{2}$% p.a. for 3 years
e $6800 at 6.2% p.a. for 6 years
f $12 500 at 7.3% p.a. for 2 years

2 Steffi invests $17 000 at 6.45% p.a. simple interest for 4 years. To what final value will her investment grow?

EXAMPLE 29

3 Calculate the simple interest for each situation.

a $2100 invested at 6% p.a. for 9 months
b $3450 invested for 5 months at 3.25% p.a.
c $10 000 invested at 6.2% p.a. for 150 days
d $4000 borrowed for 14 months at 6.55% p.a.
e $280 000 borrowed at 8.3% p.a. for 30 weeks
f $18 400 borrowed for 160 days at 14.7% p.a.

EXAMPLE 30

4 Find the annual interest rate if:

a an investment of $3000 earns $756 simple interest after 4 years
b an investment of $17 630 earns $2715.02 simple interest after 2 years
c a flat rate loan of $9000 is charged $2115 after 5 years.

EXAMPLE 31

5 For how long will $12 350 need to be invested to earn $1778.40 if the flat rate of interest is 2.4% p.a.? PS R

6 How many months will it take for an investment of $22 000 to earn $12 650 in interest if the interest rate is 6.9% p.a.?

7 How many days will it take for $15 600 to earn $436.97 in interest if the flat rate of interest is 7.1%?

8 William took 2 years to pay off a flat rate loan of $7000. His total loan repayments amounted to $8400. Calculate:

a the interest charged
b the interest rate

9 Simone and Jeff borrowed $4000 to build a garden in their backyard. They repaid the loan at $128 per month for 4 years. PS



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a Calculate the total amount they paid for the garden.

b How much interest did they pay on the loan?

c Calculate the flat rate of interest charged.

Quiz
Mental skills 3B

Mental skills 3B: Maths without calculators ANSWERS ON P. 568

Estimating square roots

We can use the square numbers to estimate square roots.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1	4	9	16	25	36	49	64	81	100	121	144

1 Study each example.

a Estimate $\sqrt{7}$.

7 is between the square numbers 4 and 9 (2^2 and 3^2 respectively).

Because 7 is closer to 9, $\sqrt{7}$ is closer to 3.

An estimate is $\sqrt{7} \approx 2.6$. (Actual answer = 2.645...)

b Estimate $\sqrt{55}$.

55 is between the square numbers 49 and 64 (7^2 and 8^2 respectively).

Since 55 is closer to 49, $\sqrt{55}$ is closer to 7.

An estimate is $\sqrt{55} \approx 7.4$. (Actual answer = 7.416...)

2 Now estimate each square root correct to one decimal place.

a $\sqrt{8}$	**b** $\sqrt{18}$	**c** $\sqrt{28}$	**d** $\sqrt{111}$
e $\sqrt{80}$	**f** $\sqrt{31}$	**g** $\sqrt{12}$	**h** $\sqrt{65}$
i $\sqrt{75}$	**j** $\sqrt{29}$	**k** $\sqrt{40}$	**l** $\sqrt{126}$

Ratios and rates

3.08

A **ratio** compares quantities of the same type measured in the same units.

It is written in the form '$a : b$' (pronounced 'a to b'), where a and b are numbers.

Skillsheet
Ratios

3.08

Interactives
Ratio and proportion
Proportion playground
Unit rates

Simplifying ratios

To **simplify a ratio**, keep dividing each term by the same number, preferably a large number such as their highest common factor (HCF), until each term is as small as possible.

Example 32

Simplify each ratio.

a 36 : 48 : 18

b 425 mL to 5 L

Solution

a $36 : 48 : 18 = \frac{36}{6} = \frac{48}{6} = \frac{18}{6}$
$= 6 : 8 : 3$

Dividing all terms by 6, their HCF

b $425 \text{ mL} : 5 \text{ L} = 425 \text{ mL} : 5000 \text{ mL}$
$= 425 : 5000$
$= \frac{425}{25} : \frac{5000}{25}$
$= 17 : 200$

Expressing both ratios in the same units

Dividing both terms by 25, their HCF

Ratio problems can be solved using **equivalent ratios** or the **unitary method**.
The unitary method requires finding one part first.

Video
Ratio problems

Example 33

During summer, the ratio of jeans to shorts sold at a beachside store was 3 : 10.
If 240 pairs of jeans were sold, how many pairs of shorts were sold?

Solution

Method 1: Equivalent ratios

Jeans : shorts $= 3 : 10 = 240 :$ _____

Since $3 \times 80 = 240$,

Number of shorts $= 10 \times 80$
$= 800$

Method 2: Unitary method

Jeans : shorts = 3 : 10

Since 240 jeans were sold, 3 parts = 240

$\therefore$ 1 part = 240 ÷ 3 — Finding 1 part first

= 80

$\therefore$ Number of shorts = 80 × 10 — Finding 10 parts

= 800

Dividing a quantity in a given ratio

- Find the total number of parts
- If using the **unitary method**, find the size of one part, then multiply to calculate each share
- If using the **fraction method**, calculate fractions of the quantity to find each share
- Check that the shares add up to the original quantity

Video
Dividing a quantity in a given ratio

Example 34

Divide a cash prize of $420 between Elyse and Henry in the ratio 3 : 4.

Solution

Method 1: Unitary method

Total number of parts = 3 + 4 = 7

$\therefore$ 7 parts = $420

1 part = $420 ÷ 7 = $60

$\therefore$ Elyse's share = 3 × $60 = $180

Henry's share = 4 × $60 = $240

Checking: $180 + $240 = $420

Method 2: Fraction method

Total number of parts = 3 + 4 = 7

$\therefore$ Elyse's share = $\frac{3}{7} \times \$420 = \$180$

Henry's share = $\frac{4}{7} \times \$420 = \$240$

Checking: $180 + $240 = $420

Simplifying rates

A **rate** compares 2 quantities of different types measured in different units.

It measures how one quantity changes with another quantity.

It is written in the form 'a/b' (pronounced 'a per b'), where a and b are units of measurement.

Example 35

Write each statement as a rate.

a Travelling 600 km in 12 hours.

b A phone call costs $6.45 for 15 minutes

Solution

a 600 km in 12 h $= 600 \div 12$
$= 50$ km/h

b \$6.45 for 15 min $=$ \$6.45 $\div 15$
$=$ \$0.43/min

Rate problems

Problems involving rates can be solved usually by multiplying or dividing.
The following strategy may help.

- Write the units of the rate x/y as a fraction: $\frac{x}{y}$
- To find the quantity in the numerator, x, **multiply** by the rate
- To find the quantity in the denominator, y, **divide** by the rate

Video Rate problems

Worksheet Rates exercise

Example 36

A car travels at 92 km/h.

a How far will it travel in $4\frac{1}{2}$ hours?

b How long will it take to travel 800 km? Answer correct to one decimal place.

Solution

a Write the units of the rate as a fraction:
km/h $= \frac{\text{km}}{\text{h}}$
To find the distance (km), multiply by the rate.
Distance $= 4\frac{1}{2} \times 92$
$= 414$ km

b To find the time (h), divide by the rate.
Time $= 800 \div 92$
$= 8.6956...$
≈ 8.7 h

EXERCISE 3.08 ANSWERS ON P. 568

Ratios and rates U F PS R

EXAMPLE 32

1 Simplify each ratio.

a	48 : 28	**b**	15 : 36	**c**	125 : 60	**d**	84 : 56
e	124 : 62	**f**	$\frac{3}{5} : \frac{1}{4}$	**g**	0.68 : 0.24	**h**	1.25 : 4.5
i	$5 : \frac{4}{5}$	**j**	$\frac{5}{8} : \frac{2}{3}$	**k**	4.8 : 0.64	**l**	15 : 18 : 6
m	27 : 36 : 9	**n**	120 : 20 : 80	**o**	$\frac{5}{6} : 3$	**p**	$2\frac{1}{2} : 5$

2 Simplify each ratio.

a	75c : \$4	**b**	5 kg : 800 g	**c**	2 hours : 45 min
d	600 mm : 2 m	**e**	8 hours : 1 day	**f**	8 mm : 4 cm
g	300 mL : 2 L	**h**	\$3.50 : 25c	**i**	6 min : 45 s

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EXAMPLE 33

3 The ratio of Ken's age to his daughter Jess' age is 10 : 3. If Ken is 30, How old is Jess?

4 Bella and Suri invest in a business in the ratio 3 : 5. If the larger investment is $520 000, find the amount of the smaller investment.

5 A survey of car buyers found that they purchased cars with colours in the ratio of white : grey : blue : other colour = 8 : 5 : 11 : 10.

A car dealer ordered 20 cars in colours other than white, grey and blue. How many white cars should be ordered?

EXAMPLE 34

6 A concrete mixture of gravel to sand to cement of 4 : 3 : 1 is needed for strong foundations. How much of each is needed to make 40 cubic metres of concrete? PS R

7 At a busy intersection involving a red-light camera, the ratio of cars running a red light to cars stopping was 1 : 79. If 90 000 cars passed through the intersection last week, how many ran the red light?



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8 Shahid and Bridie won $60 000 in a lottery. If they share the prize in the ratio 23 : 27, how much does each person receive?

9 An alloy consists of nickel and copper in the ratio 4 : 7. If the alloy weighs 3.41 kg, how much copper was used? Select the correct answer **A**, **B**, **C** or **D**.

A 0.21 kg **B** 0.85 kg **C** 1.24 kg **D** 2.17 kg

EXAMPLE 35

10 Write each statement as a rate using the units in brackets. Round answers to one decimal place if needed.

a 540 km in 8 h (km/h)
b 5 kg for $42 ($/kg)
c 2800 words in 50 min (words/min)
d 14 L for 170 km (km/L)
e 200 m in 26 s (m/s)
f 5 kg of seed for 120 m^2 (g/m^2)
g 40 days to walk 900 km (km/day)
h $215 for 3 hours ($/h)

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EXAMPLE 36

3.08

11 A truck maintains an average speed of 65 km/h. Calculate how far it will travel:

a in $2\frac{1}{2}$ h **b** from 6:30 a.m. to 3:00 p.m.

12 Walid earns $220 for an 8-hour day. PS R

a Express this pay as an hourly rate ($/h).

b How much would he earn in a 40-hour week?

c How many days will it take him to earn $1760?

13 5 kg of lamb loin chops cost $79.95.

a Express this as a rate ($/kg).

b How much would 3 kg cost?

c How many kilograms (correct to one decimal place) can be bought for $40?

14 James' heart beats at 78 beats/min.

a How many times will it beat in one hour?

b How long will it take to beat 1000 times? Answer to the nearest second.

15 Gold jewellery is classified according to its gold content. The ratio of gold to other metals is given in carats. Pure gold is 24 carats, so 10-carat gold has gold mixed with other metals in the ratio 10 : 14. PS R

a Write the ratio of gold to other metals in:

i a 9-carat bracelet **ii** an 18-carat ring

b Nami purchased a 14-carat gold necklace with a mass of 50 g. How much gold is in the necklace, correct to one decimal place?

16 A utility van has a fuel consumption of 11.6 L/100 km. Calculate how much fuel it will use for a trip of:

a 640 km **b** 58 km **c** 140 km

17 Calculate the fuel consumption, in L/100 km, of a Toyota Prius hybrid that uses 30.6 L to travel 900 km.

18 Birth rates are given as births/1000 people. A city's birth rate is 16 births/1000 people. If there were 520 births in the city last year, what was its population?

19 David claims that 5 kg of Greenie grass seed will plant an area of 1200 m^2. How much grass seed is needed for an area of 60 m^2? Select **A**, **B**, **C** or **D**.

A 0.25 kg **B** 0.4 kg **C** 1.44 kg **D** 4 kg

20 **a** Indonesia has a population of 278 million and an area of 1 919 440 km^2. Calculate its population density in persons/km^2, correct to one decimal place.

b Australia has a much smaller population of 26.6 million but a much larger area of 7 692 000 km^2. Calculate its population density, correct to one decimal place.

c If Australia was as densely populated as Indonesia, what would its population be? Answer to the nearest hundred.

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Did you know?

The beat of life

The heart of an unfit person works harder whether at rest, during activity or even while recovering from activity. A fit person's heart copes with activity better and its beat returns to normal sooner.



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	Heartbeats per minute		
	Fit	Unfit	Difference
Before activity	65	80	15
During activity	95	135	40
3 minutes after	80	115	35
6 minutes after	70	100	30
9 minutes after	65	95	30

What patterns do you notice in the differences in heart rates between fit and unfit persons?

3.09 Converting rates

It is often necessary to convert rates from one set of units to another, for example, from kilometres/hour to metres/second, or from millilitres/minute to litres/day.

Video
Converting rates 2

Example 37

Convert:

a 80 L/kg to L/g **b** 5 m/s to km/h **c** 2 mL/min to L/day

Solution

a $80 \text{ L/kg} = \frac{80}{1000} \text{ L/g}$ — 1 kg = 1000 g

$= 0.08 \text{ L/g}$

b $5 \text{ m/s} = 5 \times 3600 \text{ m/h}$ — 1 h = 3600 s

$= 18\,000 \text{ m/h}$

$= \frac{18\,000}{1000} \text{ km/h}$ — 1 km = 1000 m

$= 18 \text{ km/h}$

c $2 \text{ mL/min} = 2 \times 60 \text{ mL/h}$ — 1 h = 60 min

$= 120 \text{ mL/h}$

$= 120 \times 24 \text{ mL/day}$ — 1 day = 24 h

$= 2880 \text{ mL/day}$

$= \frac{2880}{1000} \text{ L/day}$ — 1 L = 1000 mL

$= 2.88 \text{ L/day}$

EXERCISE 3.09 ANSWERS ON P. 569

Converting rates U F PS R

EXAMPLE 37

3.09

1 Convert:

a 5 m/s to m/h
b 30 m/s to m/h
c $10/kg to $/g
d 20 mL/min to mL/h
e 40 mL/minute to mL/h
f 5 mL/s to mL/h
g 5 kg/m^2 to kg/ha
h 2.5 tonnes/day to kg/day
i $750/week to $/year
j 0.5 km/min to km/h
k 15 sheep/h to sheep/day
l $60/day to $/week

2 Convert (round to one decimal place if needed):

a 5 m/s to km/h
b 35 cm/s to m/h
c 10 cents/g to $/kg
d 40 mL/min to L/h
e 40 L/min to mL/s
f 5 mL/s to L/day
g 25 g/m^2 to kg/ha
h 2.5 tonnes/h to kg/day
i 110 km/h to m/s
j 5 kg/m to g/cm
k 8 km/L to m/mL
l 70c/min to $/h

3 Convert each speed into kilometres per hour. Round your answers to 2 decimal places.

a An African cheetah was measured running at 27 m/s.
b A German peregrine falcon dived at 97 m/s.
c A Tanzanian snake can travel at a speed of 3.3 m/s.
d A sailfish off the coast of Florida was estimated to swim 30 m/s.
e A racing cyclist rode at 23 m/s.



Stock.com/GP232

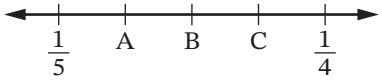
4 If Nate's reaction time is 0.9 seconds, how far will his car travel in this time if its speed is: PS R

a 60 km/h?
b 80 km/h?
c 110 km/h?

Foundation | Standard | Complex

Power plus ANSWERS ON P. 569

1 What fractions do A, B and C represent on this number line if the intervals between the fractions are equal?



2 Deepti sold a car for $14 700, making a loss of 30% on its original price. Find the original price.

3 A store marks up the price of a lounge suite by 80%. What percentage discount would bring the price of the lounge suite back to its original price?

4 Ineke went to a supermarket and bought 5 kg of potatoes. If 3 kg of potatoes cost $4.99, how much change would Ineke receive from $20?

5 2 L of water is to be shared equally between 3 people.

a Calculate how much water each person would receive (correct to 3 decimal places).

b Explain why 3 decimal places would be too accurate for practical purposes.

6 Melinda bought a pair of shoes that were reduced by $\frac{1}{3}$. If the shoes sold for $72, what was their original price?

7 After training for 6 months, Owen improved his time for running 100 m by 8%. If his new time is 10.2 seconds, what was his old time? Answer correct to one decimal place.

8 At a local football game, the ratio of men to children was 5 : 8 and the ratio of women to children was 7 : 10. What is the ratio of men to women?

9 Hannah, Nicola and Olivia share an amount of money in the ratio 2 : 3 : 5. If Nicola gives $35 to Hannah, both of them will have an equal amount of money. How much money will Olivia receive?

10 James and Jordan each drove their cars to the beach. James drove his car at a speed of 40 km/h for half the distance and at 60 km/h for the other half. Jordan drove at 50 km/h for the entire distance. Who arrived first, James or Jordan? Give reasons for your answer.

11 **a** How long will it take for an investment of $5000 to double if it is invested at 5% p.a. simple interest?

b How long will it take for an investment of $10 000 to double at the same interest rate?

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CHAPTER 3 REVIEW

Language of maths

ascending	decimal	decimal places	descending
discount	fraction	GST	integer
interest rate	irrational	loss	metres per second (m/s)
per annum (p.a.)	percentage	principal	profit
rate	ratio	rational	reciprocal
recurring	simple interest	terminating	unitary method

1 What is another name for **flat rate interest**?

2 What is the product of a fraction and its reciprocal?

3 Find a non-mathematical meaning for:

a reciprocal **b** recurring **c** terminating.

4 What is the difference between the selling price and the cost price called when the selling price is higher?

5 What does 'annum' mean, as in 'per annum'?

6 Which method involves finding the value of one part when given the value of the whole amount or several parts?

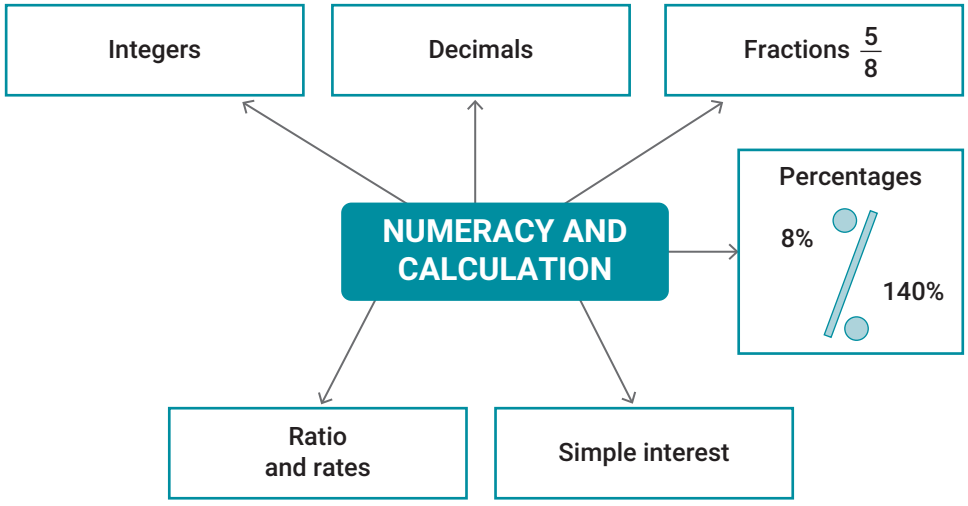
Quiz
Language of maths 3

Puzzle
Numbers crossword

Topic summary

- Write 10 questions (with solutions) that could be used in a test for this chapter. Include some questions that you have found difficult to answer.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Numeracy and calculation

Quiz
Test yourself 3

TEST YOURSELF 3

ANSWERS ON P. 569

3.01 **1** Evaluate each expression.

a $5 \times (-6) \div (-10)$ **b** $8 - (9 + 4)$ **c** $-3 \times 4 - 9$

d $5 \times 3 - 8 \times 5$ **e** $(-5)^2 - 7 \times 3$ **f** $\dfrac{4 \times -8 + (-10)}{3 \times (-2)}$

3.02 **2** Write these decimals in descending order: 0.401, 0.0483, 4.001, 0.05635.

3.02 **3** Evaluate each expression.

a $12 + 0.56$ **b** $14.5 - 6.782$ **c** 4.2×5

d 3.45×0.03 **e** $5.026 \div 4$ **f** $80.45 \div 0.05$

g 7.432×1000 **h** $345.7 \div 100$ **i** $0.605 \div 10$

j $34.1 \times 10\,000$ **k** 856.03×1000 **l** $84.6 \div 100\,000$

3.02 **4** Which expression is NOT the same as $948 \div 4$? Select the correct answer **A**, **B**, **C** or **D**.

A $9.48 \div 0.04$ **B** $94\,800 \div 400$ **C** $0.948 \div 0.004$ **D** $0.948 \div 0.4$

3.02 **5** Round 4.9948 to 3 decimal places.

3.02 **6** Convert $\frac{4}{11}$ to a recurring decimal.

3.03 **7** Which fraction is smaller, $\frac{2}{5}$ or $\frac{5}{12}$?

3.03 **8** Evaluate each expression.

a $\frac{4}{5} + \frac{3}{8}$ **b** $\frac{5}{11} - \frac{2}{3}$ **c** $2\frac{4}{7} - 1\frac{2}{5}$

d $\frac{1}{4} \div \frac{4}{5}$ **e** $\frac{4}{5}$ of \$163 **f** $\frac{1}{2} \times 2\frac{2}{3}$

g $5 \div \frac{3}{4}$ **h** $2\frac{4}{5} \div 1\frac{1}{4}$ **i** $\frac{4}{5} \div \frac{3}{4} - \frac{1}{2}$

3.04 **9** Express each percentage as a simplified fraction.

a 74% **b** 0.5% **c** $66\frac{2}{3}\%$ **d** 14.8%

3.04 **10** Express each percentage as a decimal.

a 21.4% **b** 45% **c** $8\frac{1}{3}\%$ **d** 0.2%

3.04 **11** Express each decimal and fraction as a percentage.

a 0.245 **b** $\frac{7}{8}$ **c** 1.275 **d** $\frac{19}{25}$

3.04 **12** Find:

a 42% of \$580 **b** $8\frac{1}{4}\%$ of 90 kg **c** 2.45% of \$358

3.05 **13** **a** Increase 120 L by 35%.

b Decrease \$25 by 12.5%

3.05 **14** What percentage is (correct to one decimal place):

a 5 minutes of 1 hour?

b 75c out of \$4.50?

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15 In a recent test cricket match, Steve made 286 runs in both innings, which represented 37% of the team total. How many runs did the team make in the match? 3.05

16 Calculate the selling price of a smartphone marked at $485 and discounted by 15%. Select **A**, **B**, **C** or **D**. 3.06

A $72.75 **B** $470 **C** $412.20 **D** $412.25

17 A block of land was purchased for $250 000. Four years later, it was sold for $630 000. Calculate: 3.06

a the profit

b the profit as a percentage of cost price.

18 At a sale, Garrett and Kay bought a cutlery set, saving $150 on the original price. If this discount represented 55% of the price, what was the price of the cutlery? 3.06

19 Calculate, to the nearest cent, the simple interest on: 3.07

a $8050 invested for 5 years at 4.8% p.a.

b $3890 borrowed for 7 months at 2.4% p.a.

20 Calculate the annual simple interest rate if an investment of $7000 earns $1820 in interest over 2 years. 3.07

21 Simplify each ratio. 3.08

a 200 : 450 **b** 35 mm : 2 m **c** 5 h : 45 min

22 Bill and Ben win $800 000 in Lotto. If they agree to share the prize in the ratio of 25 : 17, how much will each person receive? 3.08

23 A truck travels 581 km in 7 hours. 3.08

a Calculate its average speed in km/h.

b How far does the truck travel in 4 hours?

c How long will it take to travel 1000 km? Answer correct to one decimal place.

24 Convert: 3.09

a 0.8 mL/s to mL/h **b** 140 g/m^2 to kg/m^2

c 30 km/h to m/s **d** 345.6 m/day to cm/min

PRACTICE SET 1

ANSWERS ON P. 569

3.02 **1** Convert each fraction to a decimal and state whether it is recurring or terminating.

a $\frac{5}{8}$ **b** $\frac{7}{13}$ **c** $\frac{4}{9}$ **d** $4\frac{3}{4}$

1.03 **2** Simplify each expression.

a $7 + 5m - 9$ **b** $8p - 3m + 12p + 4m$ **c** $5n^2 - 3n - 2n^2 - 7n$

3.01 **3** Evaluate each expression.

a $7 + (-5)$ **b** $-3 - (-8)$ **c** $9 - (-4)$

d -3×5 **e** $-6 \times (-10)$ **f** $\frac{-20}{4}$

g $(-5)^2$ **h** $15 - 2 \times (-9)$ **i** $30 \div (-6) + (-3) \times 8$

j $(-2)^3 + 5 \times 4$

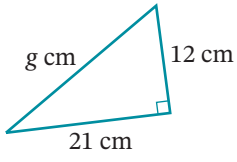
2.01 **4** Select the surds from this list of square roots.

$\sqrt{98}$ $\sqrt{9}$ $\sqrt{225}$ $\sqrt{160}$ $\sqrt{36}$ $\sqrt{52}$

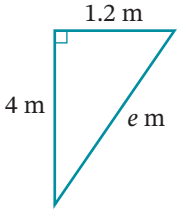
2.01 **5** Evaluate $\sqrt{(8.7)^2 - (2.6)^2}$, correct to 2 decimal places.

2.03 **6** Find, correct to one decimal place, the value of each variable.

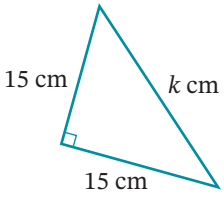
a



b



c



3.03 **7** Convert $\frac{35}{12}$ to a mixed numeral.

3.04 **8** Which of these numbers is larger than $\frac{1}{3}$? Select the correct answer **A**, **B**, **C** or **D**.

A 0.3 **B** $\frac{4}{9}$ **C** $\frac{2}{8}$ **D** 0.3333

3.02 **9** Evaluate each expression.

a 93.45×10 **b** $17.8 \div 100$ **c** 9.3×1000 **d** $234.7 \div 1000$

e $9.76 + 13.8$ **f** $15.86 + 4$ **g** $27.9 - 0.86$ **h** $12 - 8.4$

i 9.2×4 **j** 13.4×0.3 **k** $5.65 \div 0.05$ **l** $9 \div 4$

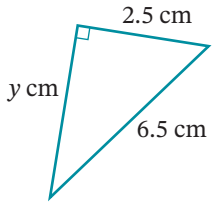
3.02 **10** Write these decimals in descending order: 5.85, 0.5867, 0.59, 5.095.

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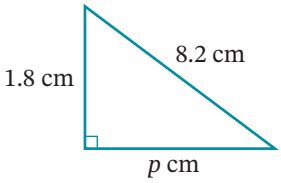
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11 Find the value of each variable. 2.04

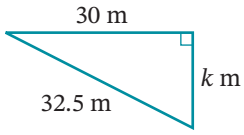
a



b



c



12 Write an algebraic expression for each statement. 1.01

a r increased by 7 **b** The difference between T and 15

c Multiply k by 4 **d** 10 divided by z

13 Evaluate each expression. 3.03

a $\frac{1}{4}+\frac{3}{5}$ **b** $\frac{7}{10}-\frac{2}{3}$ **c** $3\frac{1}{5}+1\frac{2}{3}$ **d** $4\frac{1}{2}-2\frac{4}{5}$

e $\frac{4}{7}\times\frac{3}{5}$ **f** $\frac{5}{8}\div\frac{2}{3}$ **g** $\frac{4}{5}\div 3$ **h** $1\frac{2}{7}\times 2\frac{1}{2}$

14 Arrange the fractions $\frac{2}{3}$, $\frac{3}{5}$ and $\frac{5}{8}$ in ascending order. 3.03

15 Adeeb calculated 5.64 ÷ 0.2 without using a calculator. Which of these calculations will give him the correct answer? Select **A**, **B**, **C** or **D**. 3.02

A 564 ÷ 2 **B** 56.4 ÷ 0.02 **C** 56.4 ÷ 20 **D** 56.4 ÷ 2

16 Convert each number to a percentage. 3.04

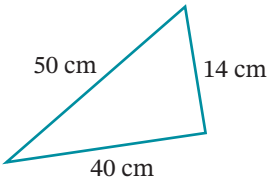
a $\frac{27}{40}$ **b** 0.27 **c** $\frac{3}{11}$ **d** 0.085

17 Calculate: 3.05

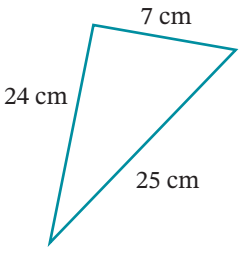
a 170% of $60 **b** 14% of $250 **c** $3\frac{1}{2}$% of $670

18 Test whether each triangle is right-angled. 2.06

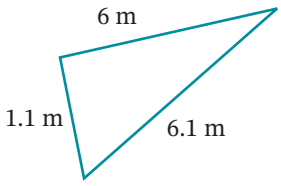
a



b



c



19 Paba bought 55 L of petrol at 189.9 cents per litre. 3.08

a How much did Paba pay (to the nearest cent)?

b How many litres can be bought for $60? Answer to one decimal place.

20 At a sale, the price of a TV was reduced by 30%. Find the original price of the TV if its sale price is $339.50. 3.06

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1.02 **21** If $a = 7$, $k = -2$ and $w = 5$, evaluate each expression.

a $a + k + w$ **b** $5kw$ **c** $\frac{a+k}{w}$

d $a^2 - w^2$ **e** $\frac{w-k}{a}$ **f** $3w + 4k$

1.04 **22** Simplify each expression.

a $2a^2 \times 4a$ **b** $5 \times 4p \times 2pd$ **c** $9b^2d \times 3bd$

1.04 **23** Simplify each expression.

a $\frac{21mn}{7n}$ **b** $36x \div 12y$ **c** $\frac{w^2v^3}{2vw}$

3.04 **24** Convert:

a 35% to a fraction in simplest form

b 0.214% to a decimal.

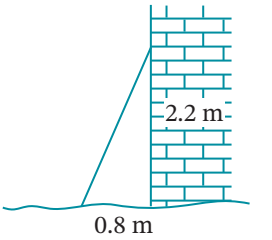
3.06 **25** A dishwasher costing $525 is sold for $699.95. Calculate the percentage profit on the cost price, correct to one decimal place.

3.08 **26** Simplify:

a 160 : 96 **b** 3 h : 45 min

c 1275 words in 15 minutes (as words/min).

2.08 **27** The foot of a ladder is placed 0.8 m from the base of a wall. If the ladder reaches 2.2 m up the wall, calculate, correct to one decimal place, the length of the ladder.



3.05 **28** **a** Increase 85 kg by 30%

b Decrease $6.40 by 18%

1.07 **29** Expand each expression.

a $8(5 - x)$ **b** $3h(2 - 7h)$ **c** $9c(2d - 3c)$

1.07 **30** Expand and simplify each expression.

a $2(5w - 3) + 4(1 + 3w)$ **b** $3(1 - 4k) - 2(2k + 1)$ **c** $4(3t - 4) - 7(1 - 2t)$

3.08 **31** Divide $450 between Klaas, David and Sarah in the ratio 3 : 5 : 7.

3.09 **32** The speed of a spacecraft leaving the Earth's gravitational pull must reach 11.2 km/s. Express this speed in km/h.

33 Simplify each expression.

1.05 **a** $\frac{9d}{11} - \frac{3d}{11}$ **b** $\frac{7k}{8} + \frac{11k}{8}$ **c** $\frac{3m}{5} + \frac{2m}{3}$ **d** $\frac{2y}{5} - \frac{3h}{4}$

1.06 **34** Simplify each expression.

a $\frac{a}{6} \div \frac{m}{5}$ **b** $\frac{2d}{w} \times \frac{5w^2}{6e}$ **c** $\frac{3x}{5} \div \frac{9x^2}{10}$ **d** $\frac{h}{4m} \times \frac{2d}{5y} \div \frac{2y}{3m}$

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35 Expand and simplify each expression. 1.07

a $4(2 + 3y - x^2) - 5(y - 1)$ **b** $-4(2 + 3y - x^2) + 5(y - 1)$

36 A rectangle has side lengths of 5 cm and 7 cm. Find the length of a diagonal as a surd. 2.05

37 A discount of \$75 represents 4% of the marked price. What is the marked price? 3.06

38 Expand each expression.

a $(p + 4)(p - 8)$ **b** $(2n + 1)(3n + 5)$ **c** $(7 - y)(7 + y)$

d $(4g - 3)^2$ **e** $(2e + 5)^2$ **f** $(3 - 2x)(x + 5)$

39 Which expression is $16mn + 24m^2$ completely factorised? Select **A**, **B**, **C** or **D**. 1.09

A $4(4n + 6m)$ **B** $8m(2n + 3m)$ **C** $4m(4n + 6m)$ **D** $m(16n + 24m)$

40 Factorise each expression. 1.08

a $3p - 15$ **b** $5(m + 4) - t(m + 4)$ **c** $-8x^2y - 12xy$ 1.08

41 Sarita spent $\frac{1}{4}$ of the day working, $\frac{1}{3}$ of the day sleeping and $\frac{1}{12}$ of the day travelling to and from work. The rest of the time was spent relaxing. 3.03

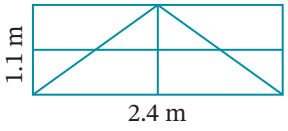
a What fraction of the day was spent relaxing?

b How many hours did Sarita spend travelling to and from work and at work?

42 Patrick bought a new computer and received a 20% discount. He paid by cash and received a further 8% discount off the discount price. What is the total percentage discount (on the original cost) that Patrick received? Select **A**, **B**, **C** or **D**. 3.06

A 8% **B** 28% **C** 23.6% **D** 26.4%

43 Calculate the length of steel required to make the frame of the gate to a paddock shown in the diagram (correct to one decimal place). 2.08



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